

# Active Inference Math Session with Octopus ~

## August 13, 2024

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Duration: 52:05 | Uploaded: Unknown Transcript method: auto\_caption

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so I'm going to share what I know well and then talk about how that might translate and there's going to be a lot of math here the goal is not to you know make it so that everything is completely you're not you won't be completely proficient in everything if this is your first time seeing this um but the goal is that maybe it highlights some connections to things that are important and can give people people some uh starting points for further exploration so when we talk about these dynamical processes there are a couple of different things a dynamical process just means something changes and a stochastic dynamical process is one where something changes in a way that is not uh completely predictable that there is some element of Randomness I'm going to start with dynamical some examples of dynamical processes that are um familiar maybe so the notation we're going to use is uh some people really like to use subscripts [Music] for these things and I do not so where other people might write  $x_{sub Z} x_{sub one} x_{sub 2} \dots$  I'm always going to write it in function notation which means the same thing this means  $X$  is something I'm measuring at time at a particular time um these are equivalent ways they're just different notations and I prefer this one but I want to say that so that if you look at a different resource and they use this subscript notation it's equivalent so um I'm going to start with discrete time because I think from a perceptual level we experience things typically as continuous time continuous space but when we're modeling it often we start with thinking of it as discrete time and then we start to think about the snapshots that we're taking as being closer and closer and a sort of an approximation of continuous time so we're going to have a discrete time and this is going to be a difference equation which is a special type of discrete time dynamical system and I'm going to start there are two things you need for one of these you need initial conditions and you need an update Rule and this is true you need these two things no matter what you are working with whether it's like an L system or a cellular automaton or a differential equation you always will want to have some type of initial or boundary condition in an update rule the initial conditions tell us

where to start and the update rule tells us from our current state how do we get to the next so the example I want to start with is the the uh dynamical or the difference equation specified by  $x_0 = 3$  that's my initial condition I'm starting with three of whatever I have at time zero and then my update rule will be  $x_{t+1}$  which this is my next next term will be given by  $2 * x_t$  which is the previous term or I shouldn't say previous I want to say now the now term so I start with three and every time step I double whatever I currently have what this does is I can use this to generate a sequence of numbers according to this rule so I would start with  $x_0$  is equal to 3 now whenever time ticks my next step  $x_1$  will be  $2 * 3$  which is six my  $x_2$  will be  $2 * \text{whatever my pre my now term is}$  so  $2 * x_1$  which is 12 and then my  $x$  value at time three will be taking the the value at time two and multiplying it by two and I'll get 24 so this process generates a a sequence of data that looks like this and I'm sure you may have seen this something like this before but I'm wanting to work from the very beginning and build up because I just think that things go more smoothly that way um are there any questions or thoughts about what we've said so far in terms of a a discrete time dynamical system and in particular having a this well I hope you're following this okay and if you're not please feel free to ask questions or if there's if you're following it and you think there are interesting ideas that emerge please feel free to bring those up as well so I could look at a similar type of rule but instead of always doing exactly the same thing I could have it where there's some type of choice that's made according to a probability distribution so let's look at the scenario of the what's called sometimes the random walk or the sometimes called the drunkard's walk in older literature where we have a person who's standing at zero initially and then every time step they flip a coin and if the if the coin is heads they move to the right and if the coin is Tails they move to the left and so we can track the trajectory of this person so I'm going to have the same information I'm going to have an initial condition um which is I'll start at  $x_0$  at time  $Z$  and then I'm going to have my update rule which tells me how it should change when the next time step happens so we're going to go left with probability  $1/2$  and we'll go right with probability  $1/2$  so this is now an example of a stochastic discrete Dynamic discrete time dynamical system because we have this additional element of Randomness that we're not always going to do the same thing and now we cannot generate a single trajectory like we did before rather instead of um and instead of a a single trajectory As Time evolves what we'll have is a probability distribution of where this walker might be so I'm going to go through this a little little thoroughly um when I start with  $x_0$  equals 0 because my my friend here is starting on on the number zero here no problem but now uh when I go to  $x_1$  I don't know whether they

flipped U left or right so there are two possibilities in terms of what it could be this is actually a probability distribution um or they could be at positive 1 and the probability that they're at positive 1 is  $1/2$  or they could be at negative one and the probability that they're at negative 1 is  $1/2$  and sometimes it's helpful to draw a tree here to show okay the this um I'll put the probabilities on the edges yes I see a hand raise so I'm gonna stop for a second sorry I just wanted to know if your screen went oh okay sorry I had trouble seeing your screen I was just trying to move it so I could see it so sorry to interrupt no no problem I totally lost I totally lost track of my own screen for a second um so I'm going to put probabilities of zero 0.5 on each of these and then when I go to x of two it becomes conditional probabilities because now there's a 0.5% chance that I move up one and I'm now at positive two um but there's also a 0.5 chance if I was at one that I move backwards to zero and similarly if I was at negative one there's a 0.5 probability that I went forward one to end up at zero and a 0.5 probability that I went backwards one to end up at -2 so now for X of two again it's a probability distribution or a random variable um where there's a 0.25 probability or 25% chance that I'm at positive 2 there's a 0.5 probability that I'm going to zero and there is a 0.25 probability that I'm at -2 and so what I'm generating here is a sequence of random variables where the next random variable uh depends on the previous one but that's the fundamental difference in a stochastic difference equation is that the differences are random and so I can't speak to a single sequence of numbers or I'm seing to speaking to a sequence of probability assignments so I'm going to stop here for questions because I kind of speed ran a whole undergraduate lecture there um but does that make sense and are you all okay with all this or have questions about this I'm tracking and good thank you cool so where it becomes a bit more challenging for me is to now pass to continuous measurements rather than discrete measurements I can do okay when the difference is allowed to be continuous so here our um our update rule was based on a discrete random variable that was only supported on plus or minus one so if I were writing this as a more formal equation like how I wrote for the first one um I could do that I could say  $x_0 = 0$  and then my update rule  $X_{t+1}$  is equal to  $X_t$  plus  $b_t$  where  $b_t$  equals positive one with probability  $1/2$  and  $b_t$  equals negative one with probability  $1/2$  and this would be a way where I'm taking this  $b_t$  random variable which is just a coin flip and now I'm writing this in my difference equation notation um and I'm putting this  $b_t$  variable for the coin flip other notation that we would commonly see is so alternate notation here would be to express this in terms of the difference so instead of expressing it this way I could also express it in terms of what is the difference between the next term in the

previous term and here I'm just subtracting this  $x$  oft term over or when we do this often people will write  $\Delta$  so another way to write this is the change at time  $T$  is equal to this coin flip random variable and so this makes it look more like a this emphasizes the fact that there's a difference here because I'm expressing this in terms of a change operator so if the symbols some people symbols are like no problem for some people they're a little a little icky but um that's all they these all mean the same thing what we could see instead is instead of the change being expressed as a coin flip the change could be drawn from something like a uh a random so what we had before was a something where the difference possibilities we might call it was a discrete random variable instead of difference possibilities just being 1 one we could see something where the Walker um takes a step again still you equ spaced but the takes a step where the length of that step is drawn from something like a normal distribution with um let's say mean equals  $z$  and standard deviation equals one and when we do that it's going to be it's still going to be possible to do this type of trajectory generation but the  $T$  they won't look like these discrete tables we will need to get into using information about sums of normal distributions and we will know that you know at time time step two our our possible positions are given by the sum of the two um identical independent normal distributions from the previous two time steps so it becomes a little bit less Concrete in terms of just writing these out um but it's it's doable and this would be something where we're looking at continuous time I'm sorry discret time changes but the possible changes are coming from a continuous random variable and then that actually creates our variables and then this is still just a sto stochastic difference equation um where it becomes a where it becomes a discrete or where it becomes a stochastic differential equation is when we shorten the time steps so that instead of there being so that we're not thinking about there being any separation but rather constantly the The Walker is constantly making a decision about whether or not to make a a very small step and over time those those instantaneous decisions add up um I don't have as good of an intuition for this but the the sort of go-to example for gaining intuition about stochastic differential equations would be to look at something like Brownie and motion where you have a particle that's constantly making you know infinites smal theoretically practically speaking microscopic random movements and over time that leads to um the fluctuation in the market or the the activity of the gas but the actual prediction of any single Market history or any single gas you can only say probabilistically and the key thing the key challenge actually is going from understanding it as a a single time step single time step single time step um to instantaneously each of these changes is happening According to some random distribution so

that's my best attempt at giving some leadup to it for people who haven't seen it before I see you say double pendulum is my go-to would you like to expand on that sorry I'm trying to unmute myself not be on video no I mean in terms of like brownie in motion like I think think about like a double pendulum also which is like you can predict the behavior with some probabilistic distribution but it's it's like there's also an element of Randomness it's not um it's not it's not fully predictable the way that a single pendulum is so there's some chaotic movement happening there like it similar to brownie Brownie and motion um is it still is it stochastic or just chaotic not sure because I um I'm saying this as a general comment it's it's deterministic so maybe it's not sarcastic so can I cross it out yes cross it out yeah because that's actually something I struggled with a lot um when I first looked at this was the difference between chaos and Randomness and something can be unpredictable long term because the complexity of the rules Aggregate and stack up but it's actually not random at all you'll always get the same answer 100 time Steps From Now with probability one cool we have an example and a non-example and both are always really helpful but again I I wouldn't um I called my friend who worked in finance and he gave me some pointers on how to build intuition on stochastic differential equations and we can continue exploring it but basically I've told you everything I know about it at this moment so can you um like just help me a little bit with the notation in the book MH um like so I I know like in equation a46 there's that random term  $\Omega$  that's like the um the the normal distribution maybe like it helps to uh augment the the difference equation or the stochastic differential equation um but so that  $\Omega$  like so that's why I to think of the double pendulum so I'm I'm thinking in the wrong place like so that  $\Omega$  to me looks like um like an interval but it's not like so but like that's why like my intuition is wrong here you know let me see I'm pulling up the textbook now a46 yeah in the appendix yeah so um it says it's a random term with a mean of zero so similar to your brownie and motion example um but like so for me I that's why I go to the double pendulum because I'm locked into that  $\Omega$  being like the interval from physics but it's not yeah it's um yeah what this is doing is saying like uh the change is the change at every you know instantaneous time step is drawn from um this normal distribution so instead of flipping a coin we're we're just sampling from like how much the particle could move left or right and the the explanation they give I was kind of trying to follow this although I used I probably could have used different notation um but they're saying if you look at a a discret a discret difference equation with um a discrete time difference equation where at every time step you pull from a distribution and then uh you start to look at the time steps getting closer and closer and closer

so it becomes instantaneous or continuous time that that's like the the actual probably rigorous way this is defined and that's also helpful information it as it it says in a rather understate way the interpretation of equation a46 is something tricky but this is saying that when you change you're going to change and whenever you change how much you change by is coming from this this normal distribution with a mean of zero and uh I'm guessing this is a standard deviation of something yeah I struggled with that upside down L term also like what what does that mean I don't know it's a it's a GMA um but I don't know what it represents here unless it's been defined somewhere it was not defined which is like something that made me crazy a bit and and then the the TOA is the time step right like in the in a47 so like some they flip back and forth between using uh  $t$  as time and  $T$  as time so it's just like a random bug in the system I think yeah I'm a little bit obsessive I'm probably going to go and search for gamma later with copy paste from this and see right so so me too and um yeah but now that like it's great that I uh I have a much better understanding of the um well hopefully if you can find gamma let me know but but already like I know that I'm I'm on I'm locked into like a non I'm locked into a deterministic system in my thought process and it's not that way so that's that was super helpful thank you I'm glad it was helpful and I'm I'm happy to for us to continue exploring it yeah like when I did differential equations it was always um or when I was studying this last time I studied this it was always in a deterministic system because we're trying to predict the outcome so yeah and my background is into discrete difference equations coming from um from either chaotic dynamical system something like the logistic equation or um I've done a lot with computer science algorithm and Analysis but all of that is also I'm I'm echoing that I'm also I'm more comfortable with deterministic and I'm also more comfortable with discrete time in terms of my intuition um Cheryl I see you have a hand up I believe is that from earlier or do you do you want to come in again my apologies no it is from earlier no worries I'm going to lower your hand if that's okay please feel free to raise it again if yeah cool um so the other thing that this is like personal request I wanted to do this because it helped me with getting the definitions um more stable and also because I am a mathematician if I can prove something um I want to and the what I wanted to prove although proof exist already in the in the textbook group documentation um was the equivalence of the different definitions of variational free energy so this will definitely get us into the weeds it's you know going to be Technical and symbolic um but I I feel it's worth doing so on page 28 in Chapter 2 equation 2.5 has a couple of different things that it defines and it just says these things are equivalent and whenever I see these things are equivalent I want to know why or at least I want

verification that they are so what we're looking at here is what's called um variational free energy this is a function that takes two arguments the first is  $Q$  which is a probability distribution of some kind and the second is  $Y$  which is um some data that we have what's important here is that what there's just one why like  $Y$  is you can think of it as an observation I believe is okay and uh this is a single piece of data I mention that because if that isn't uh if we don't recognize that  $Y$  is a single thing that we're plugging in here then some of the calculations you're like why am I allowed to do that um and I'll try to point out where that fact is important so for what's going to happen what we want to show is that couple of these things are equivalent so one of them is that we could Define this as um negative the expectation according to  $Q$  of  $x$  apologies I have I have somehow lost my PIN okay so let me switch back to this uh screen for a second so this is what we want to prove is that these three different things that are all present as the definition um or as representations of this thing are all actually the same so I'm going to call the first one version one second one version two and the third one version three and the goal would be to take each of the different versions and show how we could manipulate them so they actually look like the other one so I'm going to start with version two and show that you can manipulate it to become like version one and then I'm going to show version start with version three and show that you can manipulate it to become version one and then that will show that each of these three things are actually the same so starting with this definition of uh starting with version two if I start start with the cback lebl Divergence this complexity term and I say okay I'm starting with  $d_{KL} Q \text{ of } x$  double line  $P \text{ of } x$  minus I'm going to switch back to my whiteboard and hopefully the um hopefully The annotation will not abandon cool so I'm going to start with the the version two that we just wrote this  $d_{KL} X$  and then I'm going to take minus the  $X$  expectation over this  $Q$  distribution of the natural log of  $P \text{ of } Y \text{ given } X$  and now I'm going to take um I'm just going to plug in what the definition of these things is from the actual place where they were defined and so the definition of colc lead Divergence which I had to look up because um this would be the sum over whatever the possible  $X$ 's are of  $Q \text{ of } x$  times the natural log of  $Q \text{ of } x / p \text{ of } x$  and then so that's this part the first term the Callback lead were Divergence and then I'm going to subtract off the expectation which is the sum of  $Q \text{ of } x$  times the natural log of  $P \text{ of } Y \text{ go } X$  and now I am going to do um use properties of logarithms and I'm going to replace this part with something that it is equivalent to so this will become the sum over  $x$  and  $x$  of  $Q \text{ of } x$  times natural log of  $Q \text{ of } x$  minus natural log of  $p \text{ of } x$  and then in the interest of space um I'm also going to replace something here I'm going to replace the fact I'm going to replace this

probability  $y$  given  $x$  with the an equivalent way of rewriting it which is  $Q$  of  $x$  times natural log of and an equivalent way of writing the probability of  $Y$  given  $X$  is the probability of  $y$  and  $x$  divided by the probability of  $X$  and now I'm going to use properties of logarithm again on this part and again um the log of a ratio is equal to the difference of the logs so that will become the natural log of  $P$  of  $YX$  minus the natural log of  $P$  of  $X$  so let's write all this out I'm going to write them on two separate lines because it makes things line up a little bit nicer so this becomes  $Q$  of  $x$  this first term just stays the same and then this new term that where I'm introducing the log property becomes natural log of  $P$  The Joint distribution here minus natural log of  $P$  of  $X$  can you see what's going to happen uh there are a couple of terms that we'll cancel once we write everything out so what I'm going to do now is distribute this through so this first one becomes I'm going to take this  $Q$  of  $X$  and just distribute it through to the logs and I'm going to take this  $Q$  of  $X$  and distribute read it through to the logs so this becomes the sum over  $x$  and  $x$  of  $Q$  of  $x$  natural log of  $Q$  of  $x$  and then I can um split the summation here so this becomes the sum of  $x$  and  $x$   $q$  of  $x$  natural log  $P$  of  $X$  and then if I distribute this minus and this  $Q$  of  $x$  through this becomes minus the sum over  $x$  and  $x$   $q$  of  $x$  natural log probability of  $y$  and  $x$  and then this minus minus cancels and I get plus sum  $x$  and  $x$   $q$  of  $x$  natural log  $P$  of  $x$  and This this term is the same as this term just with signs flip so these cancel so what I'm left with here is sum over  $x$  and  $x$  of  $Q$  of  $x$  natural log  $Q$  of  $x$  minus the sum over  $x$  and  $x$  of  $Q$  of  $x$  times the natural log of  $P$  probability of  $Y$  and  $X$  it turns out that this is the definition of entropy up to a minus with uh multiply by negative one so this would be the minus of the entropy of  $Q$  of  $x$  and that this is the expectation over the  $Q$  of  $x$  distribution of the natural log of the  $Y$  so what we did was start with one way of writing this unpack what all of that actually was defined to be use some uh algebra properties of summations and logs and end up with version one of the same formula so this shows that version one and version two as we called it uh actually do represent the same thing because they can be one can be manipulated to be exactly the same as other so version one is equal to version two I'm not necessarily expecting um also I made a mistake here uh this  $e$  is not symbol for entropy it's symbol for expectation so this should be an  $H$  so here we have so if you look at equation 2 five I believe it is says these two things are equal and now we know for sure that they are and um if you if you want if it might be a fun exercise for you to play with it and see if you can get the same thing or if you agree with me but the big value that this added for me was getting more comfortable with the definition any thoughts questions or comments on that Zoom here to get the entire thing on the board so the other thing we need to do is start with version three and end up with version so



we're going to start with thank you for coming and hope you can come back next week so  
 we're going to start with version three I'm going to write the starting point which is my version  
 three here again starting with the cack Le blur Divergence and here on minus the natural law  
 log of the probability of Y so if I write out what that means I'm just going to write out the  
 formula here I'm sorry this is the probability of x given Y in this version so this is probability  
 of  $\ln$  of  $Q$  of  $x$  / by probability of x given Y and then this other part doesn't change so minus  
 the natural log of P of Y and this becomes the summation of so I'm going to do the same thing  
 I did before where I take this natural log and split it up and I'm also um that's probably enough  
 the on still okay um and it's still minus this natural  $\ln$  of P of Y term I'm going to do the same  
 thing again where I look at what is the definition of the probability of x given Y and I'm going  
 to rewrite the y as the probability of X and Y / by the Y and still minus the probability natural  
 log of probability one now we can do the same trick again where we take the log of the ratios  
 and turn it into the difference of the logs so what this becomes is some  $Q$  of  $x$  of natural log  $Q$   
 of  $x$  minus natural log um probability of X and Y and then because I'm minusing a minus this  
 becomes plus natural log of P of Y minus natural log of P of Y now if I were doing this for a  
 class I would have people who wanted to just go ahead and cancel these two natural log of P  
 of Y terms and that's not going to work so what we need to do is take this  $q$  of  $x$  and distribute  
 it through everything and while we're doing that we can go ahead and split the sums up so this  
 is going to be the sum over x and x of  $Q$  of  $x$  natural log  $Q$  of x minus the sum over x and x of  
 $Q$  of  $x$  natural log of probability of X and Y and this equation continues we have plus the sum  
 over  $Q$  of  $x$  natural log probability of Y and all of this is minus this natural log P of Y term now  
 if we were looking ahead to think about what we wanted to end up with we really just want the  
 first two terms and this sum of our x and x of  $Q$  of  $x$  natural log of P of Y is annoying so what  
 we can do is Factor this in front and now I'm going to erase and rewrite for interest of space so  
 what this becomes is natural log of P of Y times the sum over x and x of  $Q$  of  $x$  because this  
 natural log of P of Y was just a constant so we canull it in front and then the key Insight here is  
 that this n this sum over x and X of  $Q$  of  $x$  is equal to one since  $Q$  of  $x$  is a distribution so I can  
 take the summation and just turn it into multiply by one which means that this term and this  
 term actually do cancel so when I take this natural log P of Y minus natural log of P of Y I just  
 delete both of those because cancel and here again I'm left with the negative entropy of  $Q$  of  $x$   
 minus the expected value over the  $Q$  of  $x$  distribution of the natural log of P of XY which is  
 equal to version one of the variational frey formula so mathematically all three ways of  
 writing this are actually the same and algebraically and you can rewrite them so that they work

out to be starting with one you can get to the other and um that's a cool cool thing that I enjoyed doing but also doing that helped me internalize and make more definite some of the definitions of the formulas and to get an understanding of things like why is it negative entropy instead of positive why is it important it why only be a single observation there are other other um versions of the proofs that are well exp explained that are available in the Koda so um I'd encourage you to look at those as well but mainly it's just about I think the proof is worth doing for multiple reasons I do need to go soon or do you all have any thoughts or questions before we dismiss and um if you have any longer thoughts or questions let's discuss them in Discord and bring them back to this meeting next week cool well thanks for coming out everyone and I'll see yall uh in a week